

Team Burnout & Morale Early Warning Radar

A Non-Invasive Engineering Resilience & Early Fatigue Detection System for PulsePM

DOCUMENT VERSION	TARGET PLATFORM	ENGINEERS & REVIEWERS	DATE & LIFECYCLE
v1.0.0 (Production Blueprint)	PulsePM Client & Services	Senior Full-Stack Engineering	September 2026 — Active

1 Executive Overview & Problem Statement

In high-velocity engineering teams, burnout and morale degradation are chronic compounding issues rather than overnight events. Traditional agile project management tools (such as Jira, Linear, or Asana) only capture failure retrospectively: when deliverables are missed, sprint commitments fail, or contributors resign.

The **Team Burnout & Morale Early Warning Radar** is a proactive early warning radar designed to give Project Managers (PMs) actionable, empathetic foresight into team strain **10 to 14 days before** missed deadlines or employee turnover occur.

Non-Invasive Architecture Guarantee: The radar introduces zero surveillance software, zero screen tracking, and zero background microphone listening. It operates entirely as an algorithmic aggregator over existing platform metadata: task transition logs, weekend submission timestamps, and daily standup progress notes.

2 The 4 Core Algorithmic Signals

The radar synthesizes four normalized dimensions (0 to 100) extracted from existing platform data:

SIGNAL DIMENSION	DATA ORIGIN & FORMULA	BURNOUT CORRELATION RATIONALE
1. Task Churn Index (TCI)	Ratio of status oscillation between In Progress, Blocked, and To Do: TCI = min(100, (State_Flips / Active_Days) * 40)	Frequent context switching and re-work deplete cognitive stamina without delivering finished value.
2. Overdue & Rollover Load (ORA)	Days past end_date + sprint boundary rollover count: ORA = min(100, (Overdue_Days * 8) +	Carrying stale tickets from sprint to sprint produces psychological fatigue and feelings of falling behind.

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	(Rollover_Count * 15))	
3. Off-Hours & Weekend Activity (OWR)	Logs submitted on Sat/Sun or outside 08:30–19:30 local time: $\text{OWR} = \min(100, (\text{Weekend_Logs} / \text{Total_Logs}) * 200)$	Direct indicator of workload spillover into personal recovery time, destroying work-life balance.
4. Blocker & Sentiment Friction (BSF)	Proportion of has_worked === 0 + negative friction keywords: $\text{BSF} = (\text{Blocker_Days_Pct} * 0.7) + (\text{Keyword_Friction} * 0.3)$	External dependencies (e.g. staging certs, vendor sandboxes, missing credentials) create feelings of helplessness.

3 Composite Scoring Model & Risk Tiers

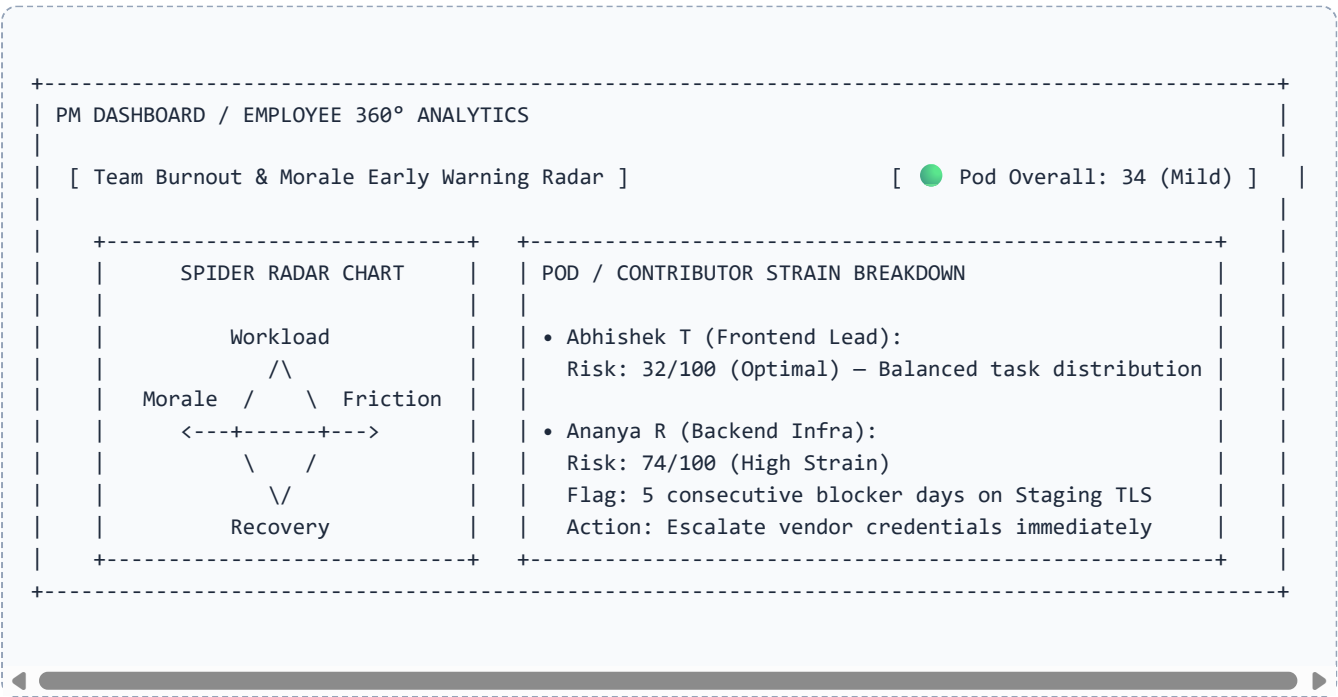
The composite **Burnout Risk Score (BRS)** is computed using an empirically weighted linear combination:

$$\text{BRS} = (\text{TCI} * 0.25) + (\text{ORA} * 0.25) + (\text{OWR} * 0.25) + (\text{BSF} * 0.25)$$

SCORE	RISK TIER	OBSERVED TEAM STATE	RECOMMENDED PM ACTION
0 – 30	OPTIMAL FLOW	Steady progress, low churn, consistent working hours.	Maintain current cadence; protect focus blocks.
31 – 60	MODERATE STRAIN	Occasional late logs, 1–2 blockers, moderate context switching.	Check in during 1-on-1s; verify ticket requirements.
61 – 80	ELEVATED FATIGUE	Repeated overdue deliverables, weekend activity spikes.	Deprioritize 1–2 tickets; rebalance pod workloads.
81 – 100	CRITICAL BURNOUT	Severe multi-day blocker stall, chronic off-hours logging.	Immediate leadership escalation; clear external blocker.

4 Component Architecture & UI/UX Placement

To ensure zero backend disruption, the feature is structured as a client-side calculation and presentation layer consuming already-fetched task and daily log data:



5 Implementation Plan & Rollout Phases

Strict Safety Constraint: Zero modifications to backend databases, SQLite tables, or existing controller APIs. The radar computes its telemetry purely on existing datasets delivered via `/projects/all/tasks` and `/employees/:id/analytics`.

PHASE	TARGET FILE / SCOPE	IMPLEMENTATION DELIVERABLES
Phase 1	<code>client/src/utils/burnoutRadarCalculator.js</code>	Pure computation engine. Takes raw tasks & logs, outputs normalized dimensions (TCI, ORA, OWR, BSF, BRS) and tailored PM action bullet points.
Phase 2	<code>client/src/components/BurnoutRadarModal.jsx</code>	Self-contained modal / card with pure SVG polygon radar chart (no heavy D3/Chart.js dependency), dark/light mode parity, and contributor breakdown.
Phase 3	<code>Employee360View.jsx</code> & <code>OtherWorkspaces.jsx</code>	Adds a designated "Burnout & Morale Radar" entry button in PM views to open the radar drawer/modal, preserving all existing KPI cards and tables.
Phase 4	Vite Build & Smoke Test	Execute <code>npm run build</code> , verify zero bundle regressions, test backdrop-blur styling and dark mode contrast.

6 Ethical & Psychological Safety Safeguards

No Punitive Metrics: The score is never labeled as developer competence or velocity. It is explicitly labeled as *"Process & External Friction Risk"*.

Blameless Problem Solving: Alerts emphasize external blockers (e.g. *"Blocked by missing staging DB credentials"*) to direct management focus onto removing impediments.

Restricted Visibility: Radar details are restricted to PM and Superuser management views, avoiding cross-contributor comparisons.